TurboGator

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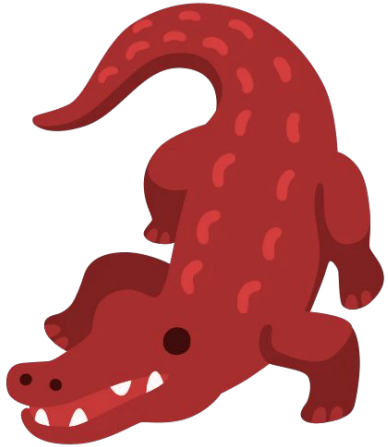

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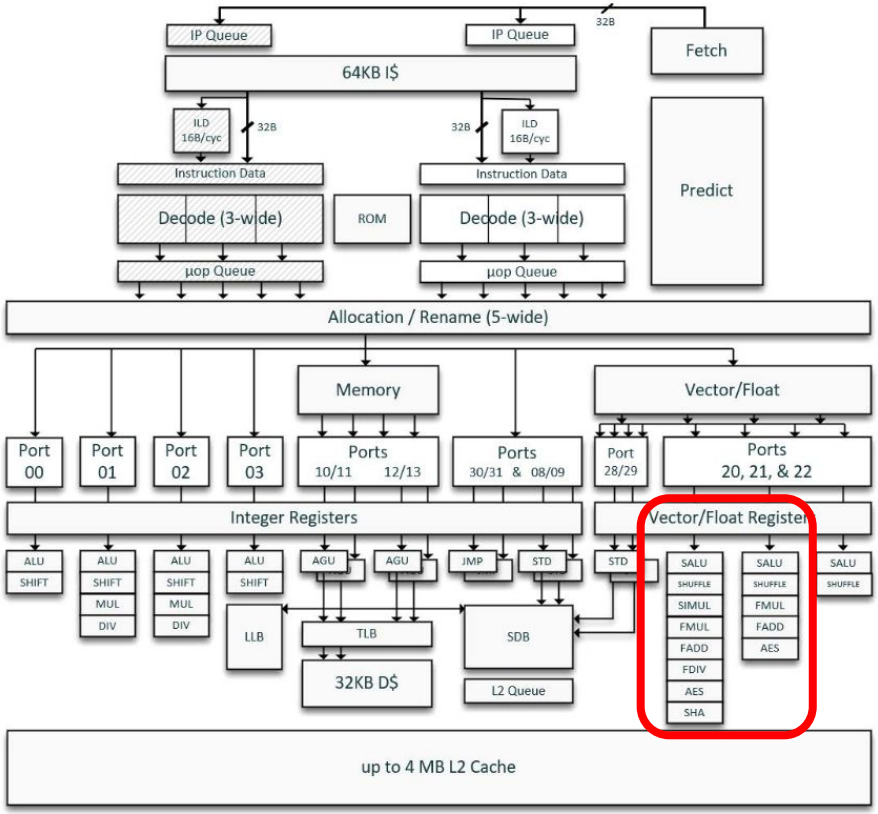
Todo Slides:

- ⬛ Bottlenecks fully identified
- ⬛ Bottlenecks optimized
  - Sparsity
  - Simplicity
  - Memory Layout Optimizations
    - Flattening & Contiguous vectors
    - Blocking ?
  - Others ?
- ⬛ Memory layout optimization
- ⬛ Runtime & roofline plots
- ⬛ (BOLD) TODO Vectorization

Recap Server setup



- Intel N100 Alder Lake E core (Gracemont)
- Intel SSE4.1, Intel SSE4.2, Intel AVX2, FMA3
- TurboBoost off 800MHz, 12 GB RAM
- 256b instructions get "cracked" in two 128b instruction

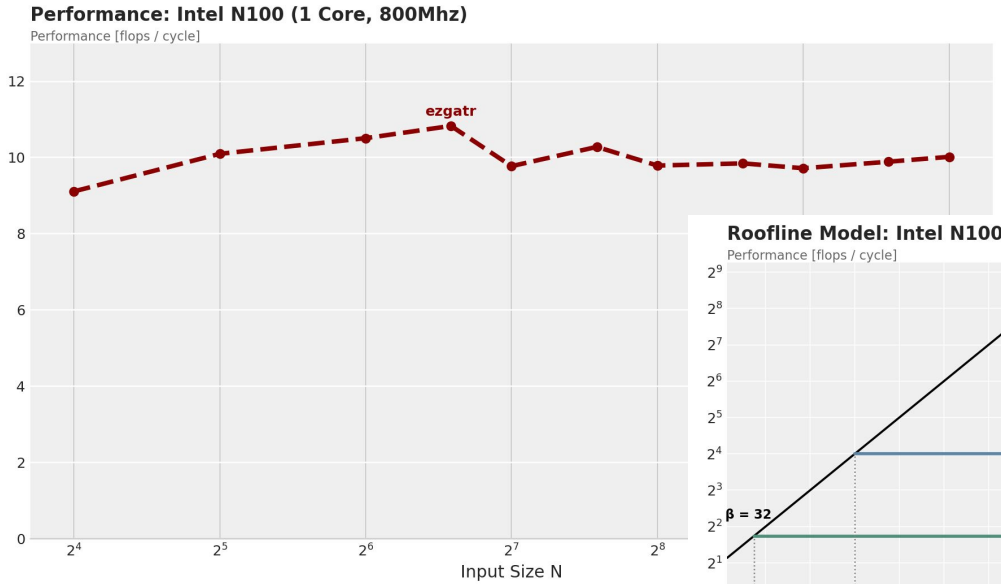
π_s = 3.33 flops/cycle

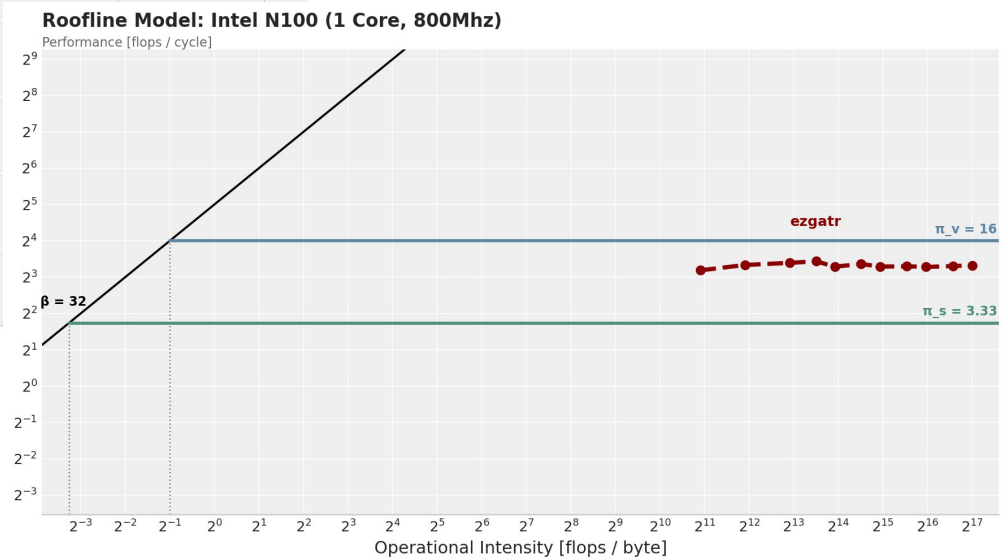
π_v = 16 flops/cycle

β = 32 bytes/cycle

L1d = 32 KiB; Latency 3 Cycles

Last time on TurboGator…





New sweep

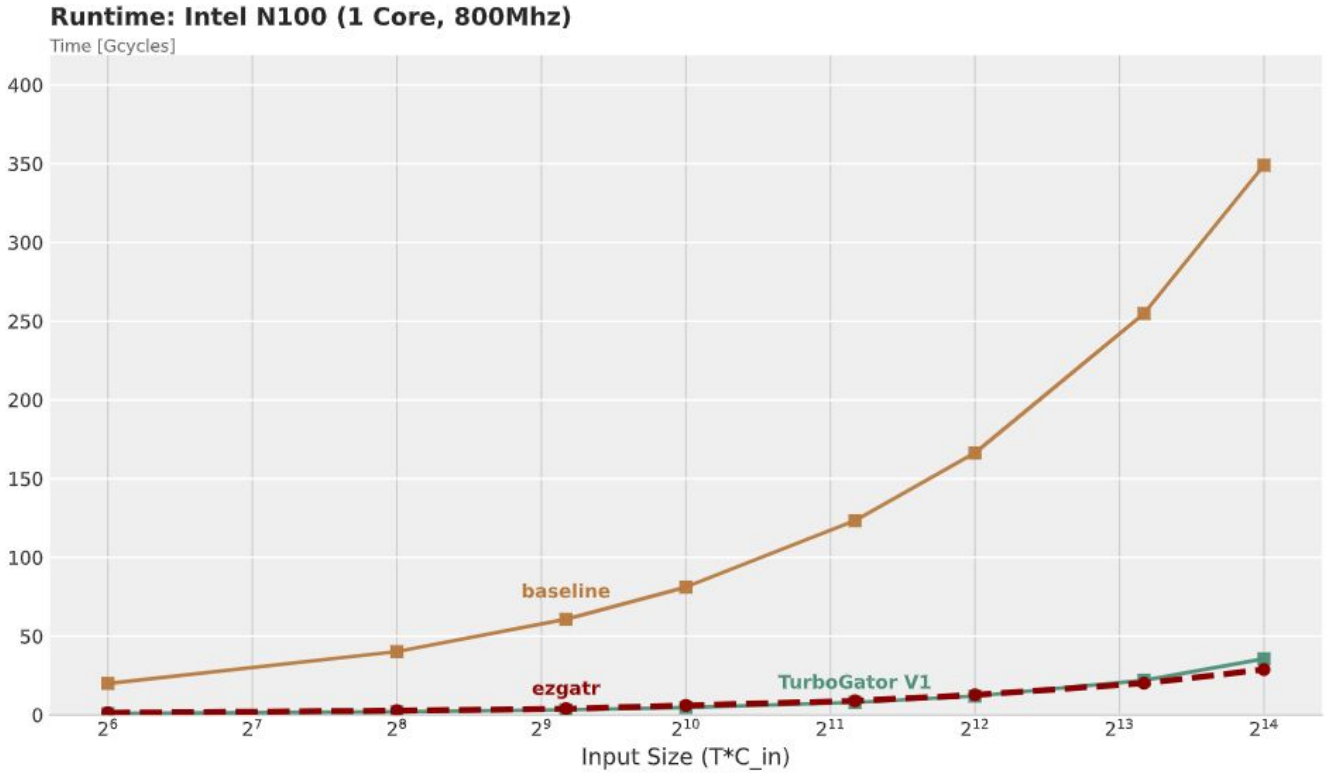
X = (8, T, C_in, 16)

T = 32 * N

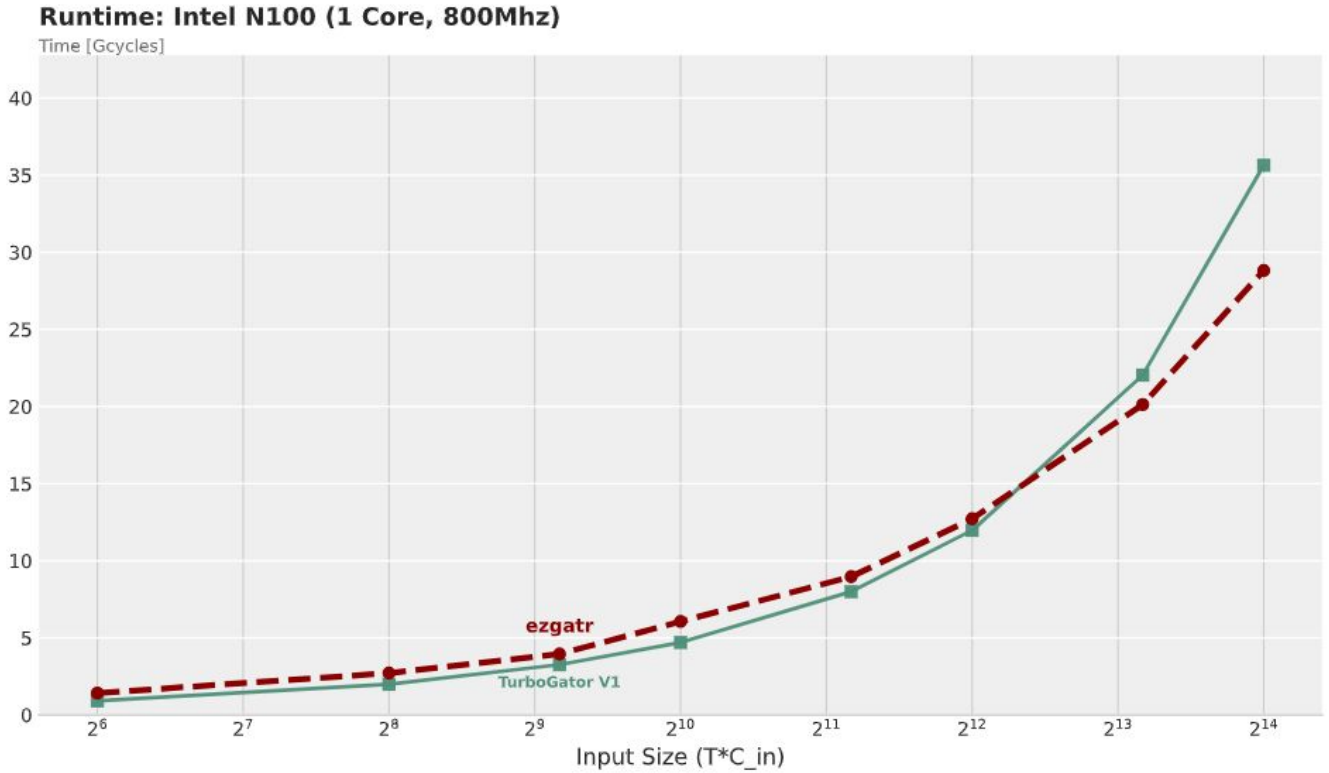
C_in = 2 * N

N=1 -> (32, 2)	= 64
N=2 -> (64, 4)	= 256
N=3 -> (96, 6)	= 576
N=4 -> (128, 8)	= 1024
N=6 -> (192, 12)	= 2304
N=8 -> (256, 16)	= 4096
N=12 -> (384, 24)	= 9216
N=16 -> (512, 32)	= 16384

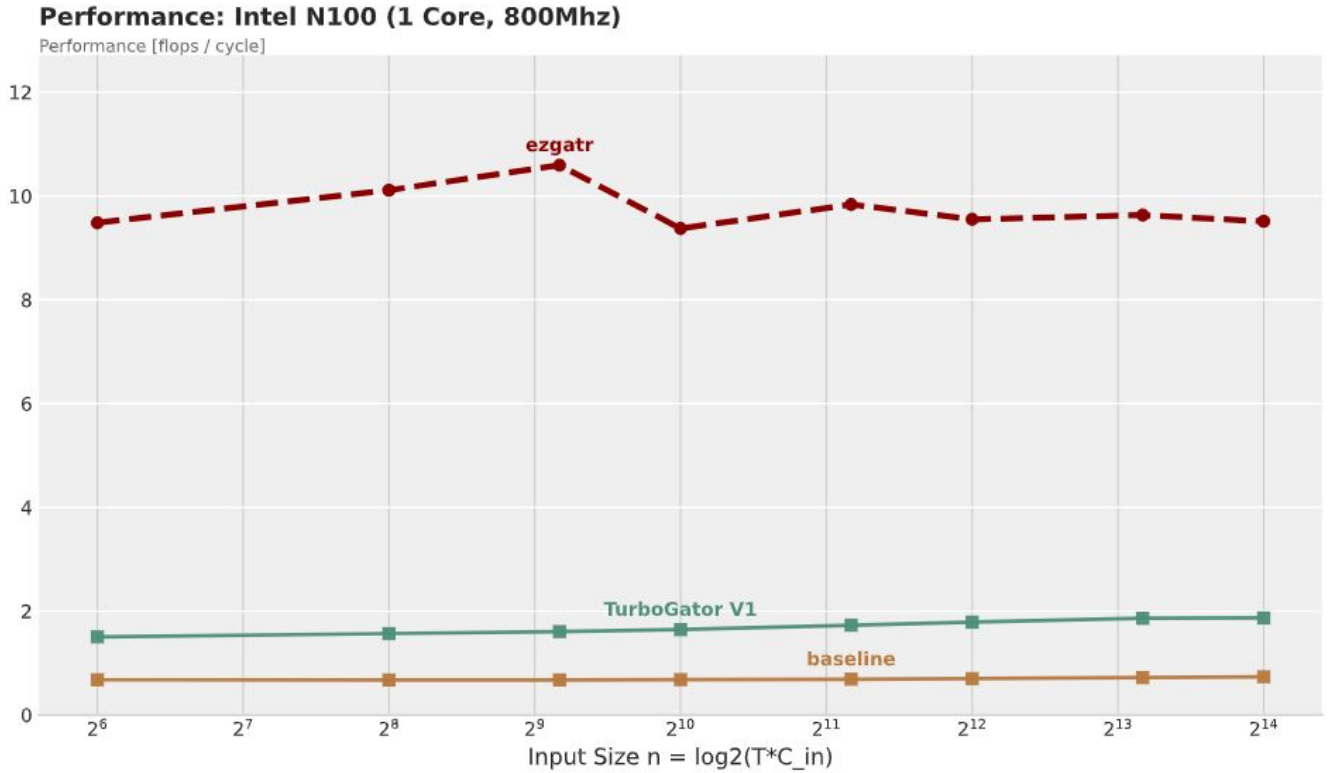
Runtime Today



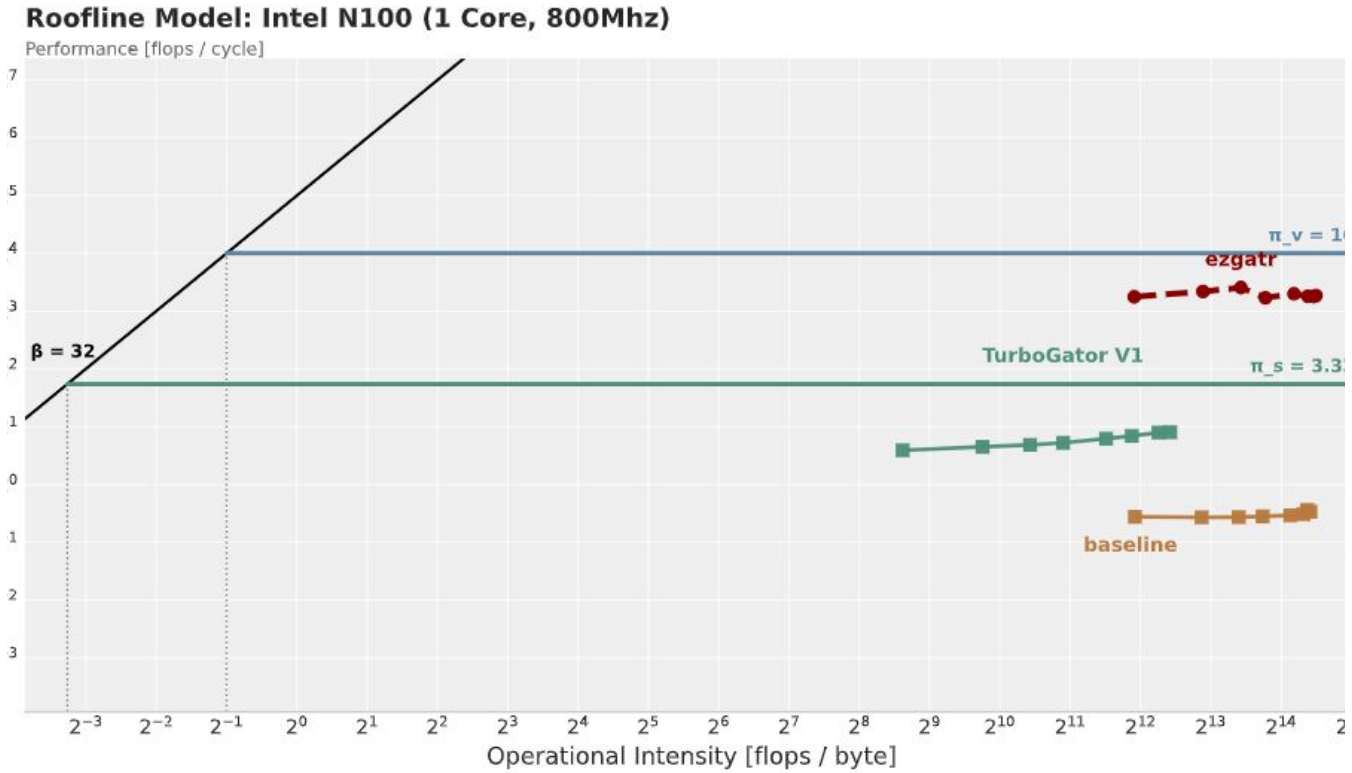
Runtime Today (ignore baseline)



Performance Today

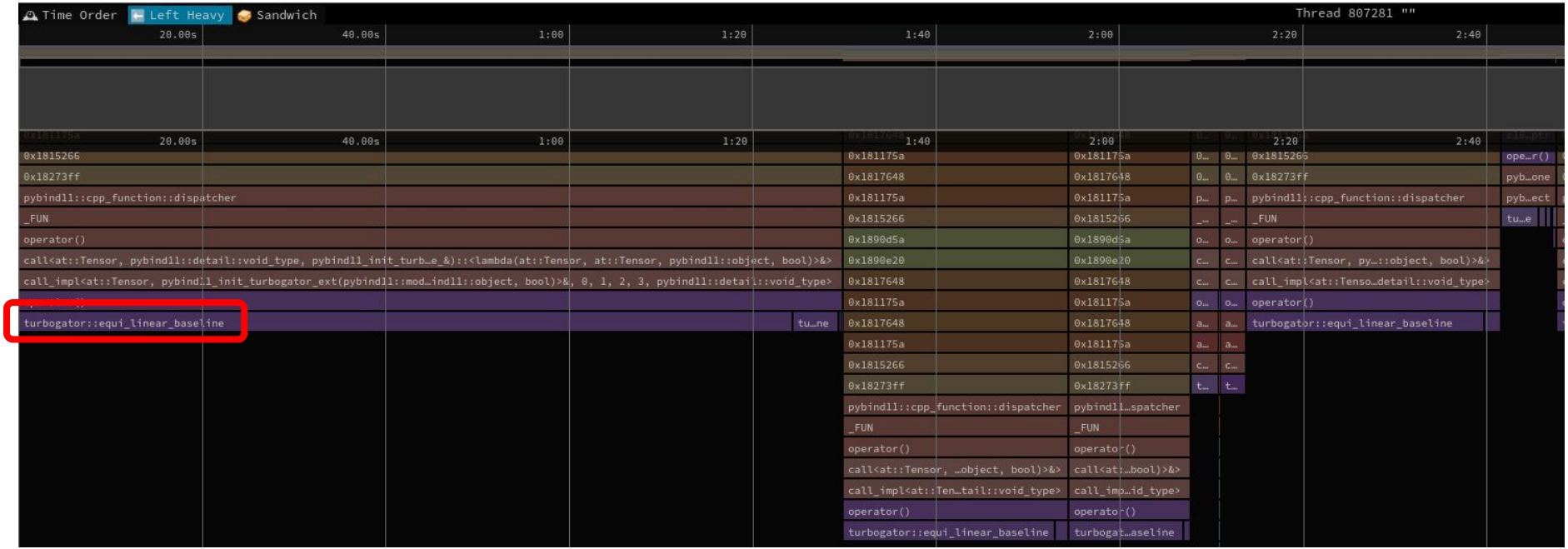


Roofline Today



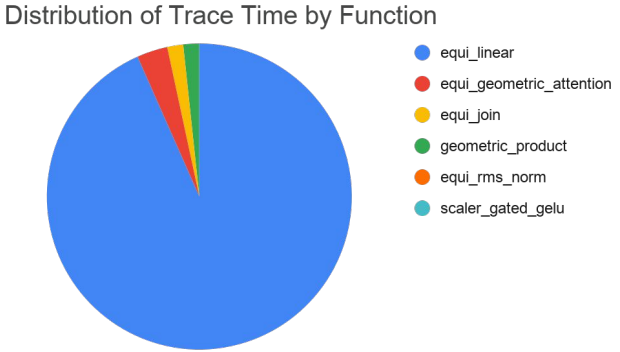
Profiling (pytorch Profiler)

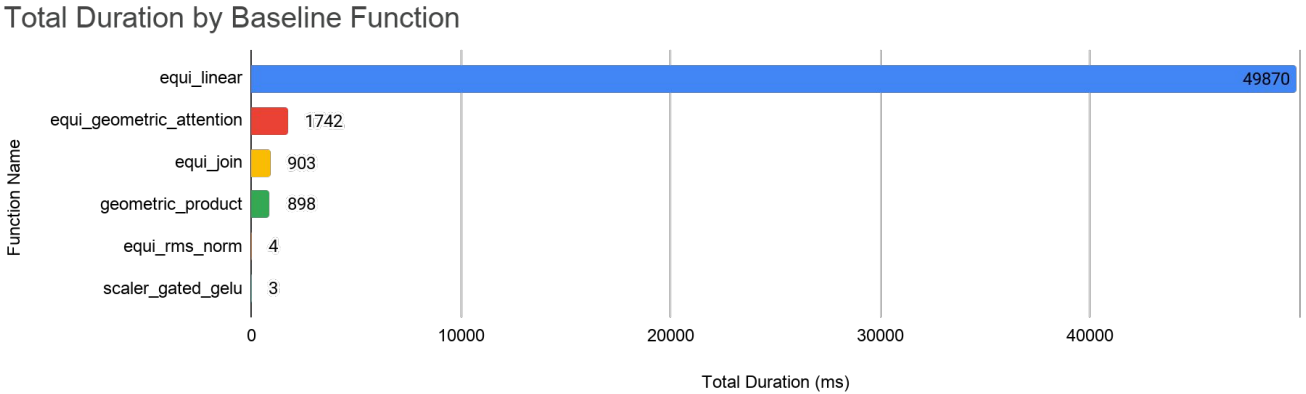
Profiling (py-spy)



Bottlenecks identified

Function Name	Call Count	Total Duration (ms)	% of Total Trace
equi_linear	22	49870	92
equi_geometric_attention	4	1742	3
equi_join	4	903	1
geometric_product	4	898	1
equi_rms_norm	8	4	0.007
scaler_gated_gelu	4	3	0.006





Bottleneck Optimizations

- ⬛ Simplicity
- ⬛ Scalar Replacement
- ⬛ Sparsity
- ⬛ Memory Layout Optimizations
  - Flattening & Contiguous vectors
  - Blocking
- ⬛ Vectorization with AVX (coming soon)

Simplicity

```
for (int i = 0; i < 4; ++i)
    for (int j = 0; j < 4; ++j)
        for (int k = 0; k < N_DAA; ++k)
            out[k] += basis[i][j][k] * r[i] * r[j];
```



```
    out[0] = -(r[3]*r[3]);
    out[1] = -(r[0]*r[0] + r[1]*r[1] + r[2]*r[2]);
    out[2] = 2.f * r[0]*r[3];
    out[3] = 2.f * r[1]*r[3];
    out[4] = 2.f * r[2]*r[3];
```

Scalar replacements

```
// step 2: out[b,o,d] = sum_{i,s} kernel[o,i,d,s] * x[b,i,s]
for (size_t b = 0; b < batch; ++b)
{
    for (size_t oc = 0; oc < out_channels; ++oc)
    {
        for (size_t d = 0; d < 16; ++d)
        {
            float sum = 0.0f;
            for (size_t ic = 0; ic < in_channels; ++ic)
            {
                for (size_t s = 0; s < 16; ++s)
                {
                    sum += kernel[((oc * in_channels + ic) * 16 + d) * 16 + s]
                    * x[b * in_channels * 16 + ic * 16 + s];
                }
            }
            out[b * out_channels * 16 + oc * 16 + d] = sum;
        }
    }
}
```

```
// step 2: out[b,o,d] = sum_{ic,s} kernel[o,ic,s,d] * x[b,ic,s]
// OPT1:
// - register accumulator: acc[16] holds all d values for one (b, oc)
// - x[b,ic,s] is loaded once per (ic,s) and reused across all 16 d
for (size_t b = 0; b < batch; ++b)
{
    for (size_t oc = 0; oc < out_channels; ++oc)
    {
        float acc[16] = {0.0f};

        for (size_t ic = 0; ic < in_channels; ++ic)
        {
            for (size_t s = 0; s < 16; ++s)
            {
                float xv = x[b * in_channels * 16 + ic * 16 + s];
                for (size_t d = 0; d < 16; ++d)
                {
                    acc[d] += kernel[((oc * in_channels + ic) * 16 + s) * 16 + d] * xv;
                }
            }
        }

        for (size_t d = 0; d < 16; ++d)
        {
            out[b * out_channels * 16 + oc * 16 + d] = acc[d];
        }
    }
}
```

Sparsity

```
float data[16][16][16] = {};
LocalGPBasis()
{
    data[0][0][0] = 1.0f;
    data[0][2][2] = 1.0f;
    data[0][3][3] = 1.0f;
    data[0][4][4] = 1.0f;
    data[0][8][8] = -1.0f;
    data[0][9][9] = -1.0f;
    data[0][10][10] = -1.0f;
    data[0][14][14] = -1.0f;
    data[1][0][1] = 1.0f;
```

...

95% reduction 😭

Sparsity II

```
float unnorm_basis[9][16][16] = {};
float norm_basis[9][16][16] = {};

LocalEquiLinearBasis()
{
    const std::vector<std::vector<BasisElement>> basis_elements = {
        {0},
        {1, 2, 3, 4},
        {5, 6, 7, 8, 9, 10},
        {11, 12, 13, 14},
        {15},
        {std::pair<int, int>{1, 0}},
        {std::pair<int, int>{5, 2}, std::pair<int, int>{6, 3}, std::pair<int, int>{7, 4}},
        {std::pair<int, int>{11, 8}, std::pair<int, int>{12, 9}, std::pair<int, int>{13, 10}},
        {std::pair<int, int>{15, 14}},
    };
```

Sparsity II

```
for (size_t b = 0; b < batch; ++b)
{
    for (size_t oc = 0; oc < out_channels; ++oc)
    {
        float acc[16] = {0.0f};

        for (size_t ic = 0; ic < in_channels; ++ic)
        {
            for (size_t s = 0; s < 16; ++s)
            {
                float xv = x[b * in_channels * 16 + ic * 16 + s];
                for (size_t d = 0; d < 16; ++d)
                {
                    acc[d] += kernel[((oc * in_channels + ic) * 16 + s) * 16 + d] * xv;
                }
            }
        }

        for (size_t d = 0; d < 16; ++d)
        {
            out[b * out_channels * 16 + oc * 16 + d] = acc[d];
        }
    }
}
```

256 → 26 FMAs per (b, oc, ic)

```
for (size_t ic = 0; ic < in_channels; ++ic)
{
    const float *w_oi = w_use.data() + (oc * in_channels + ic) * 9;
    const float *x_bi = x + (b * in_channels + ic) * 16;

    // load weights for (oc, ic)
    const float w0 = w_oi[0];
    const float w1 = w_oi[1];
    const float w2 = w_oi[2];
    const float w3 = w_oi[3];
    const float w4 = w_oi[4];
    const float w5 = w_oi[5];
    const float w6 = w_oi[6];
    const float w7 = w_oi[7];
    const float w8 = w_oi[8];

    // w=0
    acc[0] += w0 * x_bi[0];

    // w=1
    acc[1] += w1 * x_bi[1];
    acc[2] += w1 * x_bi[2];
    acc[3] += w1 * x_bi[3];
    acc[4] += w1 * x_bi[4];

    // w=2
    acc[5] += w2 * x_bi[5];
    acc[6] += w2 * x_bi[6];
    acc[7] += w2 * x_bi[7];
    acc[8] += w2 * x_bi[8];
    acc[9] += w2 * x_bi[9];
    acc[10] += w2 * x_bi[10];
```

...

Memory Layout - Flatting & Cont. Vec

- Flattened Q, K vector projections and ensured contiguous memory access.
- Together with blocking significantly improves cache efficiency for GEMM score computation.

Memory Layout - Blocking

- Currently only implemented in equi_geom_attn
  - Around 2x runtime improvement, more efficient caching
- Coming for equi_linear in V1.1
  - We expect massive gains, since the runtime share of equi_linear is much larger than equi_geom_attn

Future considerations

- Vectorisation
- Automatic FLOPs and Data Movement
- Intel VTune didn't work, perf flags missing (counters missing from silicon)
- We have to change server infra…